

Sound Reactive Disco Light Box

Designed by VoltEdge

Datasheet



1. General Description

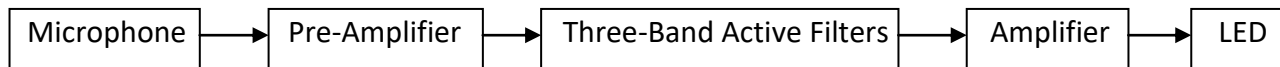
The Sound Reactive Disco Light Box is designed to convert audio signals into a visual light display using analog signal processing techniques. The system captures ambient sound through a microphone, amplifies and filters the signal into multiple frequency bands, and drives LED arrays accordingly. The brightness variation of the LEDs represents the intensity and frequency content of the input sound signal in real time.

The system operates using a battery-powered supply and does not rely on digital processing, enabling fast response and simple implementation.

2. Key Features

- Fully analog sound processing system
- Three-band frequency separation for audio visualization
- Smooth LED brightness variation proportional to sound intensity
- Battery-powered operation
- Regulated low-voltage supply for stable IC operation
- Simple and cost-effective design

3. System Block Diagram



Block Description:

The microphone converts sound waves into an electrical signal. This signal is amplified, filtered into low, mid, and high frequency components, and used to control LED groups corresponding to each frequency band.

4. Functional Description

4.1 Power Supply Section

The system is powered by a 9 V DC battery. A 78L05 linear regulator is used to generate a stable 5 V supply for all integrated circuits. The ICL7660S is also used in the power supply section to support analog signal conditioning. This arrangement ensures stable and reliable operation of the system.

4.2 Pre-Amplifier Stage

An electret microphone is used to sense ambient sound and convert it into a low-level electrical signal. Since the microphone output amplitude is small, a pre-amplifier stage is implemented using the LM358N operational amplifier. This stage amplifies the audio signal to a suitable level for further processing while preserving the original signal characteristics.

4.3 Filter and Signal Processing Stage

The amplified audio signal is applied to three active filter circuits implemented using the NE5532P operational amplifier. These filters separate the audio signal into three distinct frequency bands, allowing different portions of the sound spectrum to be processed independently.

The frequency ranges of the filters are designed as follows:

- **Low-frequency band:** 0 Hz to 200 Hz
- **Mid-frequency band:** 250 Hz to 3 kHz
- **High-frequency band:** Above 3 kHz

Each filtered output provides a continuous analog signal whose amplitude corresponds to the strength of that frequency range in the input sound.

4.4 Amplifier and LED Drive Stage

The outputs of the filter circuits are further amplified and applied directly to the LED drive stage. The LEDs respond continuously to the amplitude of the filtered signals, producing smooth brightness variations. This approach allows accurate visualization of both sound intensity and frequency distribution without discrete switching behavior.

4.5 LED Output Section

The output stage consists of a total of 30 LEDs arranged into three color-coded bands. Each band contains 10 LEDs and is driven by one of the filtered frequency ranges.

- **Low-frequency band:** 10 LEDs (Color 1)
- **Mid-frequency band:** 10 LEDs (Color 2)
- **High-frequency band:** 10 LEDs (Color 3)

This arrangement enables clear visual differentiation of low, mid, and high frequency components present in the audio signal.

5. Electrical Specifications

Parameter	Rating			Unit
	Min.	Typ.	Max.	
Supply Voltage (V_S)	8.0	9.0	9.6	V
Regulated Voltage (V_{REG})	4.9	5.0	5.1	V
Total Supply Current (I_S)	40	70	100	mA
Audio Input Voltage ($V_{IN, RMS}$)	0.02	0.15	0.5	V
Audio Input Current ($I_{IN, RMS}$)	0.1	1	5	μ A
Input Impedance (Z_{IN})	1	3	—	M Ω
Filter Output Voltage (RMS)	0.2	1.5	2.5	V
LED Drive Voltage (RMS)	1.8	2.2	3.0	V
LED Drive Current (I_{OUT})	5	10	20	mA
Operating Temperature (T_A)	0	25	70	$^{\circ}$ C

6. Mechanical Specifications

Parameter		Rating	Unit
Overall Length		150	mm
Overall Width		150	mm
Overall Height		55	mm
PCB Dimensions	Length	97	mm
	Width	60	mm
Total Weight		360	g
Enclosure Material		PLA (Polylactic acid)	—
Mounting Type		Enclosed, free-standing	—

7. Components Used

- Electret microphone
- LM358N operational amplifier
- NE5532P operational amplifier
- ICL7660S switched-capacitor voltage converter IC
- 78L05 Linear Regulator
- LEDs (30 total, three different colors)
- Resistors and capacitors
- 9 V DC battery

8. Advantages and Limitations

Advantages

- Real-time response to audio signals
- Smooth and accurate LED brightness variation
- Simple circuit architecture
- Low-cost implementation

Limitations

- Performance affected by ambient noise
- Fixed frequency band ranges
- Limited customization compared to digital systems

9. Conclusion

The Analog Sound Reactive Disco Light Box demonstrates effective use of analog amplification, filtering, and signal-driven output control to convert audio signals into a visual display. The project highlights practical implementation of multi-band audio processing using operational amplifiers and provides an intuitive representation of sound characteristics.

The complete circuit schematic and PCB layout are provided in the appendices.

Appendices

Appendix A – Circuit Schematic

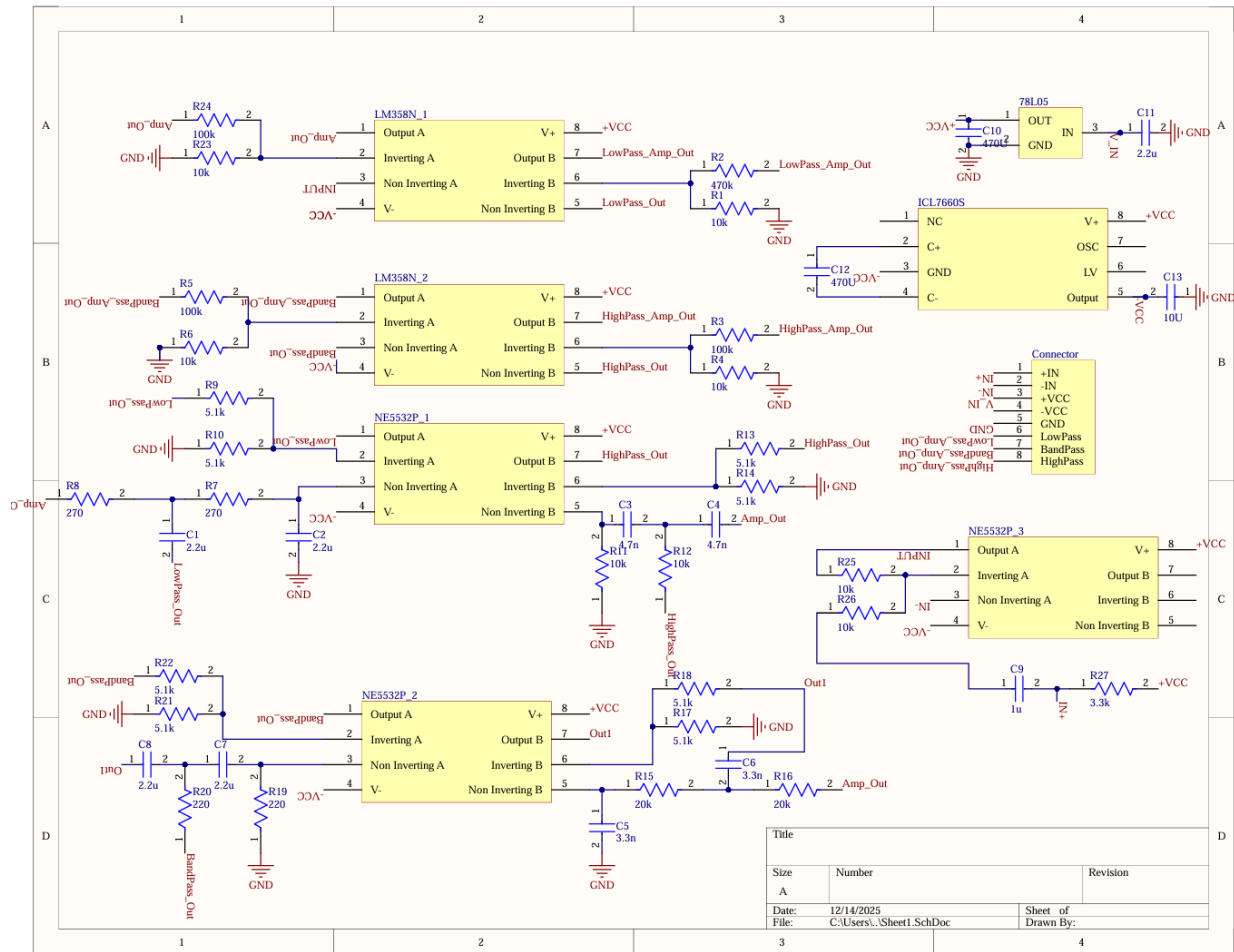


Figure A1: Complete circuit schematic of the Analog Sound Reactive Light System

Appendix B – PCB Layout

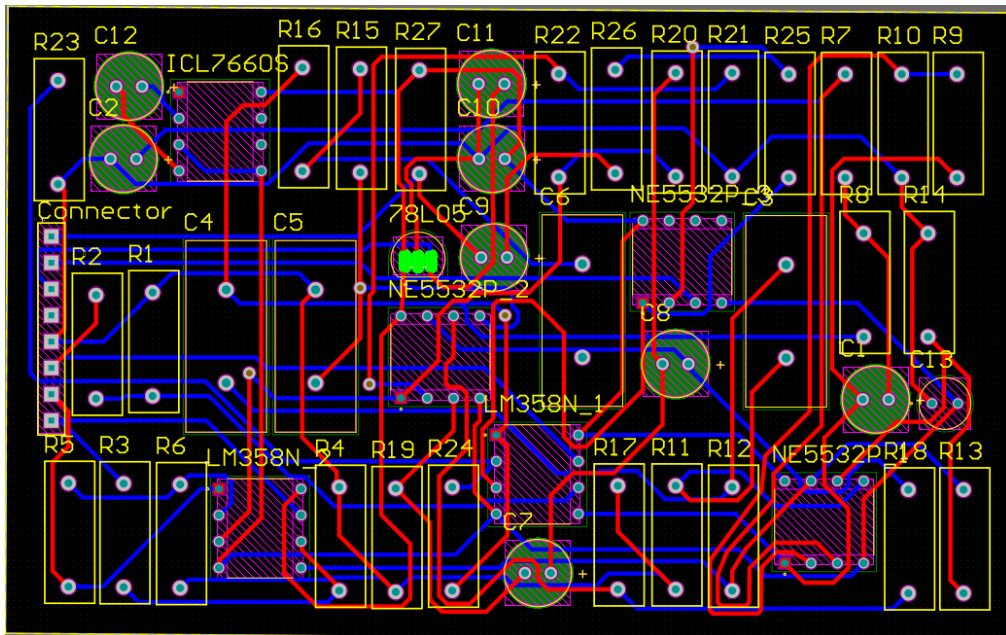


Figure B1: 2D PCB layout of the two-layer board showing top and bottom layer connections

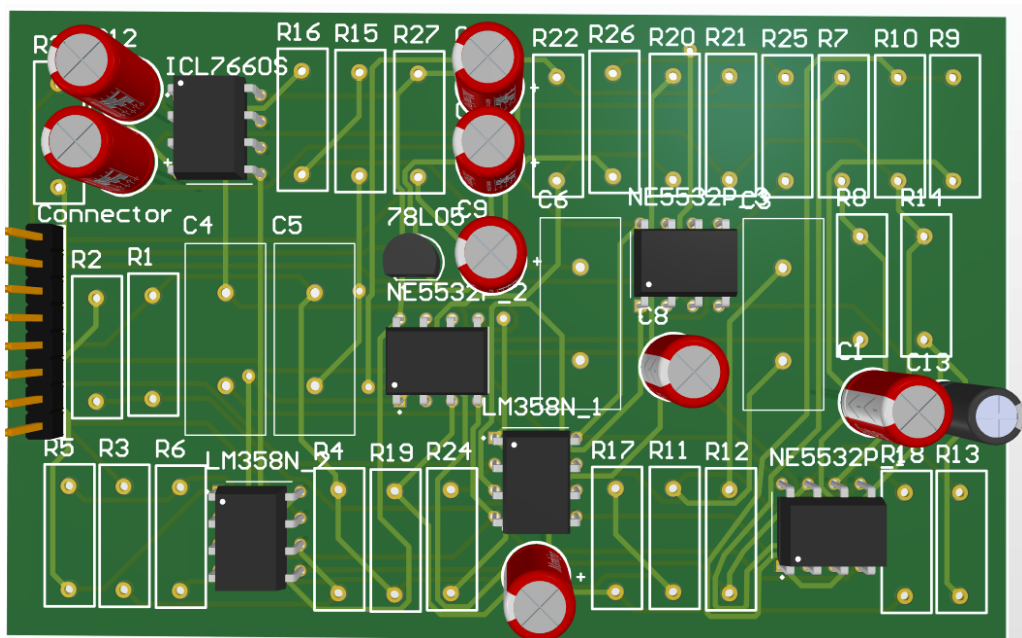


Figure B2: 3D view of the assembled two-layer PCB showing component placement

Appendix C – Enclosure Design

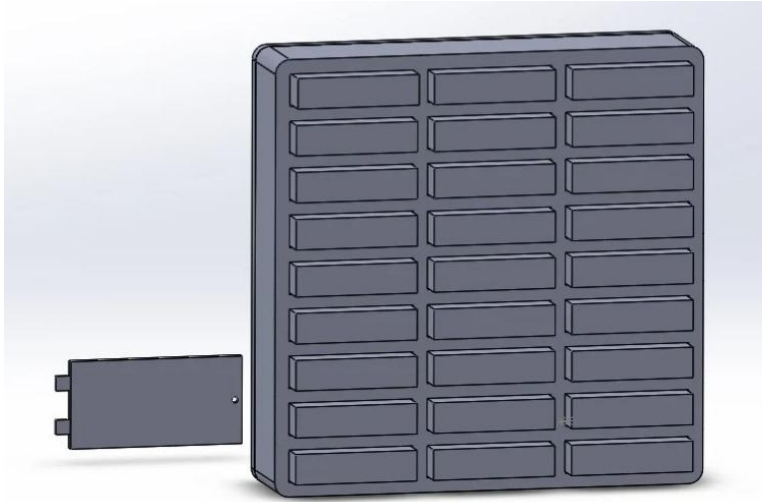


Figure C1: Top view of the enclosure designed using SolidWorks



Figure C2: Top view of the fabricated enclosure showing the final implementation

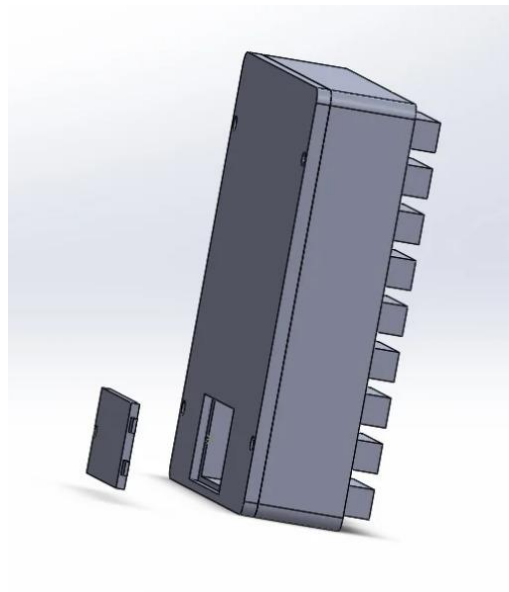


Figure C3: Side view of the enclosure designed using SolidWorks